In Chapter 7, we will provide a test of equality for two population variances. However, for the above data, the computed sample standard deviations are approximately equal considering the small sample sizes. Thus, the required conditions necessary to construct a confidence interval on $\mu_1 - \mu_2$—that is, normality, equal variances, and independent random samples—appear to be satisfied. The estimate of the common standard deviation σ is

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{9(.3234)^2 + 9(.2406)^2}{18}} = .285$$

The t-value based on df $= n_1 + n_2 - 2 = 18$ and $a = .025$ is 2.101. A 95% confidence interval for the difference in mean potencies is

$$(10.37 - 9.83) \pm 2.101(.285)\sqrt{1/10 + 1/10}$$

$$.54 \pm .268 \text{ or } (.272, .808)$$

We estimate that the difference in mean potencies for the bottles from the production line and those stored for 1 year, $\mu_1 - \mu_2$, lies in the interval .272 to .808. Company officials would then have to evaluate whether a decrease in mean potency of size between .272 and .808 would have a practical impact on the useful potency of the drug.

EXAMPLE 6.2

A school district decided that the number of students attending their high school was nearly unmanageable, so it was split into two districts, with District 1 students going to the old high school and District 2 students going to a newly constructed building. A group of parents became concerned with how the two districts were constructed relative to income levels. A study was thus conducted to determine whether persons in suburban District 1 have a different mean income from those in District 2. A random sample of 20 homeowners was taken in District 1. Although 20 homeowners were to be interviewed in District 2 also, one person refused to provide the information requested, even though the researcher promised to keep the interview confidential. Thus, only 19 observations were obtained from District 2. The data, recorded in thousands of dollars, produced sample means and variances as shown in Table 6.3. Use these data to construct a 95% confidence interval for $(\mu_1 - \mu_2)$.

TABLE 6.3
Income data for Example 6.2

	District 1	District 2
Sample Size	20	19
Sample Mean	18.27	16.78
Sample Variance	8.74	6.58

Solution A preliminary analysis using histograms plotted for the two samples suggests that the two populations are mound-shaped (near normal). Also, the sample variances are very similar. The difference in the sample means is

$$\bar{y}_1 - \bar{y}_2 = 18.27 - 16.78 = 1.49$$

The estimate of the common standard deviation σ is

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{19(8.74) + 18(6.58)}{20 + 19 - 2}} = 2.77$$

The t-percentile for $a = \alpha/2 = .025$ and df $= 20 + 19 - 2 = 37$ is not listed in Table 2 of the Appendix, but taking the labeled value for the nearest df less than 37 (df $= 35$), we have $t_{.025} = 2.030$. A 95% confidence interval for the difference in mean incomes for the two districts is of the form

$$\bar{y}_1 - \bar{y}_2 \pm t_{\alpha/2} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

Substituting into the formula, we obtain

$$1.49 \pm 2.030(2.77)\sqrt{\frac{1}{20} + \frac{1}{19}} \qquad \text{or} \qquad 1.49 \pm 1.80$$

Thus, we estimate the difference in mean incomes to lie somewhere in the interval from $-.31$ to 3.29. If we multiply these limits by \$1,000, the confidence interval for the difference in mean incomes is $-$\$310 to \$3,290. This interval includes both positive and negative values for $\mu_1 - \mu_2$, so we are unable to determine whether the mean income for District 1 is larger or smaller than the mean income for District 2.

We can also test a hypothesis about the difference between two population means. As with any test procedure, we begin by specifying a research hypothesis for the difference in population means. Thus, we might, for example, specify that the difference $\mu_1 - \mu_2$ is greater than some value D_0. (*Note:* D_0 will often be 0.) The entire test procedure is summarized here.

A Statistical Test for $\mu_1 - \mu_2$, Independent Samples

The assumptions under which the test will be valid are the same as were required for constructing the confidence interval on $\mu_1 - \mu_2$: population distributions are normal with equal variances and the two random samples are independent.

H_0: 1. $\mu_1 - \mu_2 \leq D_0$ (D_0 is a specified value, often 0)
2. $\mu_1 - \mu_2 \geq D_0$
3. $\mu_1 - \mu_2 = D_0$

H_a: 1. $\mu_1 - \mu_2 > D_0$
2. $\mu_1 - \mu_2 < D_0$
3. $\mu_1 - \mu_2 \neq D_0$

T.S.: $$t = \frac{(\bar{y}_1 - \bar{y}_2) - D_0}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

R.R.: For a level α, Type I error rate and with df $= n_1 + n_2 - 2$,
1. Reject H_0 if $t \geq t_\alpha$.
2. Reject H_0 if $t \leq -t_\alpha$.
3. Reject H_0 if $|t| \geq t_{\alpha/2}$.

Check assumptions and draw conclusions.